Pedestrian Prediction

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Pedestrian Prediction

Data Load

This section prepares the final modelling dataset based on the EDA results: numeric formatting issues are corrected, and the categorical variables `abs_mb_category_simple` and `education_low_year_3cat` are created. The table below shows the structure of the resulting dataset, including each variable's name, class, and example values.

Table 1: Structure of model_df

Variable	Class	Example
obs_peak_cross_ped_per_h	numeric	0.333333333333333, 4, 47
obs_peak_along_ped_per_h	numeric	12.3333333333333, 7.66666666666667, 101.333333333333
commercial_buildings	numeric	0, 0, 6
parking_areas	numeric	0, 1, 3
overture_food_drink_count_500m	numeric	0, 3, 19
overture_grocery_convenience_pharmacy_count_500m	numeric	0, 0, 8
overture_other_shopping_count_500m	numeric	0, 19, 35
footpath_both_sides	factor	FALSE, TRUE, TRUE
overture_shopping_mall_present_500m	factor	FALSE, FALSE, TRUE
park_bushland_reserve	factor	TRUE, TRUE, TRUE
abs_mb_category_simple	factor	Residential, Other, Other
education_low_year_3cat	factor	KIN, KIN, Y_plus

Linear Model

Variable Selection using Step Function

This step applies stepwise selection (both directions) to `obs_peak_along_ped_per_h`, starting from two models: an empty model and a model with all EDA-selected predictors. Comparing the results from both starting points helps identify a stable and consistent set of predictors for the linear model.

```
summary(linear_step_both_null)
```

Model from Null Start

```
##
## Call:
## lm(formula = obs_peak_along_ped_per_h ~ commercial_buildings +
##     parking_areas + education_low_year_3cat, data = model_df)
```

```
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -93.56 -21.26  -3.47  12.79 154.03
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     21.053     18.151   1.160   0.2523
## commercial_buildings            23.460      3.093   7.586  1.6e-09 ***
## parking_areas                  -18.176      8.113  -2.240   0.0302 *
## education_low_year_3catKIN     -26.485     16.734  -1.583   0.1206
## education_low_year_3catY_plus   15.730     23.281   0.676   0.5028
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 42.45 on 44 degrees of freedom
## Multiple R-squared:  0.6679, Adjusted R-squared:  0.6378
## F-statistic: 22.13 on 4 and 44 DF,  p-value: 4.595e-10
```

commercial_buildings, parking_areas, and education_low_year_3cat were selected. commercial_buildings has a strong, significant positive effect, consistent with the EDA finding that commercial density is the strongest predictor of pedestrian activity. parking_areas is significant, but its negative coefficient contradicts the positive relationship found in the EDA, suggesting possible confounding. Neither level of education_low_year_3cat is significant. The model explains about 64% of the variance (adjusted R^2 = 0.638).

```
summary(linear_step_both_full)
```

Model from Full Start

```
##
## Call:
## lm(formula = obs_peak_along_ped_per_h ~ commercial_buildings +
##     parking_areas + overture_food_drink_count_500m + overture_shopping_mall_present_500m +
##     abs_mb_category_simple + education_low_year_3cat, data = model_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -80.605 -20.066   0.557  18.386 116.179
##
## Coefficients:
##                                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)                             23.761     17.810   1.334   0.1897
## commercial_buildings                    21.340      3.651   5.845  7.8e-07
## parking_areas                          -17.796      8.190  -2.173   0.0358
## overture_food_drink_count_500m           3.217      1.538   2.091   0.0429
## overture_shopping_mall_present_500mTRUE -33.729     18.190  -1.854   0.0711
## abs_mb_category_simpleCommercial        -38.001     19.906  -1.909   0.0635
## abs_mb_category_simpleOther               5.931     14.146   0.419   0.6773
## education_low_year_3catKIN              -32.984     16.628  -1.984   0.0542
## education_low_year_3catY_plus             8.922     22.693   0.393   0.6963
##
## (Intercept)
## commercial_buildings                   ***
```

```
## parking_areas                           *
## overture_food_drink_count_500m          *
## overture_shopping_mall_present_500mTRUE .
## abs_mb_category_simpleCommercial        .
## abs_mb_category_simpleOther
## education_low_year_3catKIN              .
## education_low_year_3catY_plus
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 40.43 on 40 degrees of freedom
## Multiple R-squared:  0.7261, Adjusted R-squared:  0.6713
## F-statistic: 13.25 on 8 and 40 DF,  p-value: 4.045e-09
```

Four more predictors were retained: `overture_food_drink_count_500m`, `overture_shopping_mall_present_500m`, and `abs_mb_category_simple`, alongside `education_low_year_3cat`. `commercial_buildings` and `parking_areas` remain significant with the same effect directions. `overture_food_drink_count_500m` is significant and positive, consistent with the EDA. The remaining added predictors are only marginally significant (p between 0.05 and 0.09). This model explains slightly more variance (adjusted $R^2 = 0.671$), but the improvement is modest relative to the added complexity.

```
anova(linear_step_both_null, linear_step_both_full)
```

Comparison of Stepwise Models

```
## Analysis of Variance Table
##
## Model 1: obs_peak_along_ped_per_h ~ commercial_buildings + parking_areas +
##     education_low_year_3cat
## Model 2: obs_peak_along_ped_per_h ~ commercial_buildings + parking_areas +
##     overture_food_drink_count_500m + overture_shopping_mall_present_500m +
##     abs_mb_category_simple + education_low_year_3cat
##   Res.Df   RSS Df Sum of Sq      F  Pr(>F)
## 1     44 79276
## 2     40 65395  4     13881 2.1227 0.09581 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
AIC(linear_step_both_null, linear_step_both_full)
```

```
##                         df      AIC
## linear_step_both_null    6 513.1105
## linear_step_both_full   10 511.6783
```

A partial F-test shows no significant improvement from the additional predictors ($F = 2.12$, $p = 0.096$), and the AIC values are nearly identical (513.1 vs. 511.7). The simpler model (`commercial_buildings`, `parking_areas`, `education_low_year_3cat`) is therefore preferred, as it achieves a comparable fit with fewer predictors.

Linear Final Model

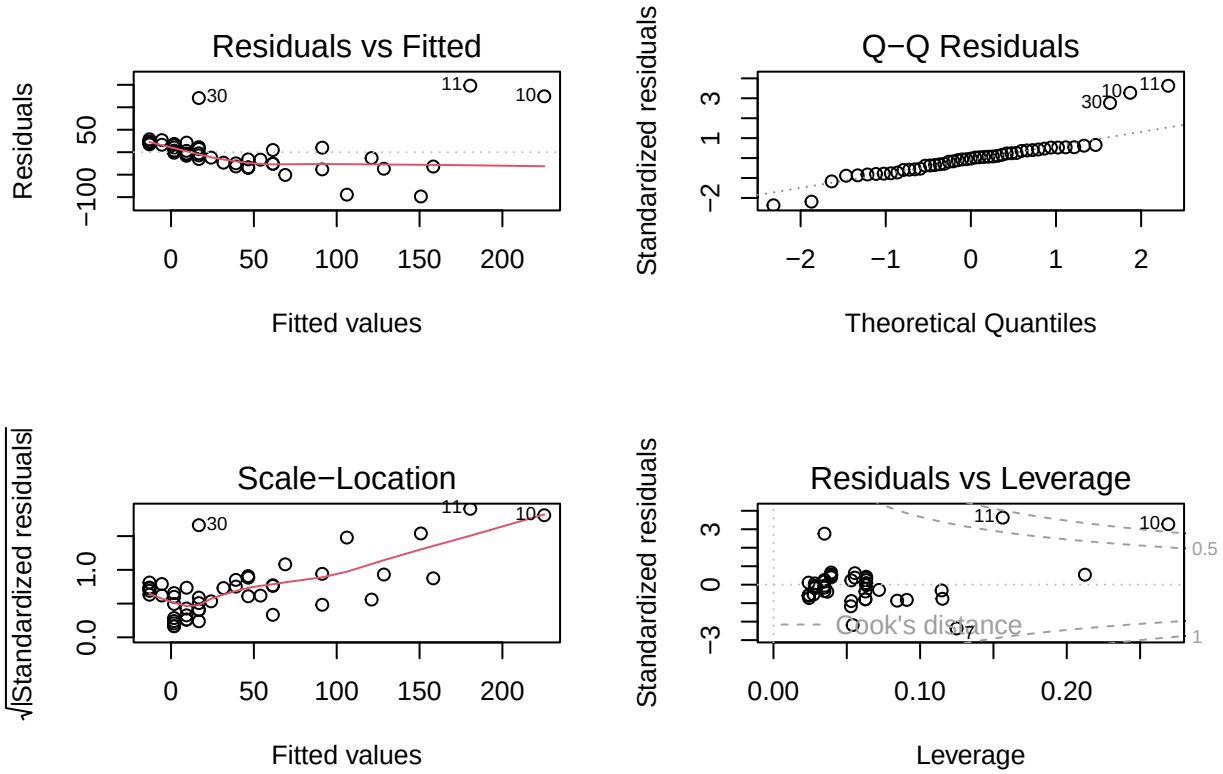
Based on the stepwise comparison, `education_low_year_3cat` was excluded, as neither level reached significance. The final linear model retains only `commercial_buildings` and `parking_areas`.

```
### Linear final model
```

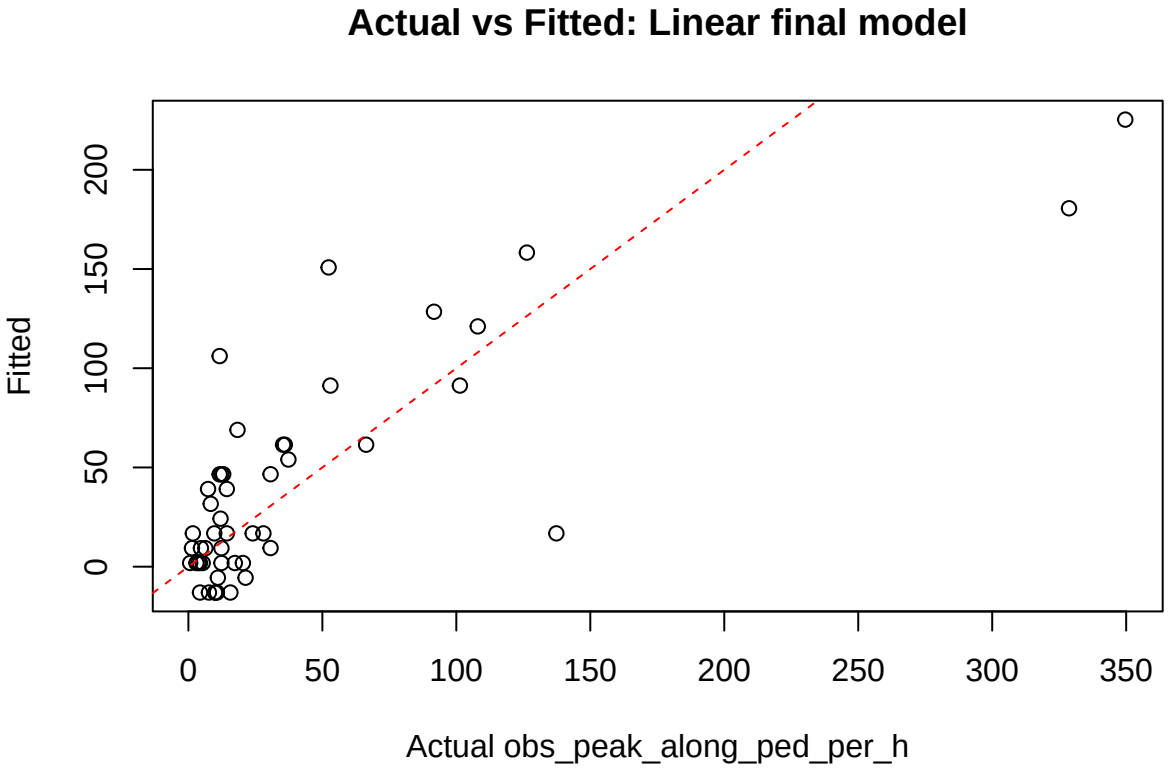
```r
linear_final_model <- lm(obs_peak_along_ped_per_h ~ commercial_buildings + parking_areas,
                         data = model_df)
summary(linear_final_model)
```

```
##
## Call:
## lm(formula = obs_peak_along_ped_per_h ~ commercial_buildings +
##     parking_areas, data = model_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -98.471 -24.787  -1.177  16.553 148.056
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)             1.844     11.174   0.165   0.8697
## commercial_buildings   22.337      3.175   7.036 8.06e-09 ***
## parking_areas         -14.867      8.371  -1.776   0.0824 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 44.44 on 46 degrees of freedom
## Multiple R-squared:  0.6195, Adjusted R-squared:  0.603
## F-statistic: 37.45 on 2 and 46 DF,  p-value: 2.227e-10
```

`commercial_buildings` remains strongly significant with a positive effect, consistent with the EDA. `parking_areas` is marginally significant ($p = 0.082$), and its negative coefficient still contradicts the positive relationship observed in the EDA, indicating likely confounding. The model explains about 60% of the variance (adjusted $R^2 = 0.603$), slightly lower than the earlier three-predictor model, but with a simpler and more interpretable structure. Multicollinearity among the predictors of all three final models is checked with variance inflation factors in the Appendix (Multicollinearity Check).



The diagnostic plots show clear problems with this model. The Residuals vs Fitted and Scale-Location plots both show increasing spread at higher fitted values, indicating heteroscedasticity. The Q-Q plot shows departure from normality in the upper tail, with points 10, 11, and 30 as notable outliers. Point 11 also shows relatively high leverage in the Residuals vs Leverage plot, suggesting it has a disproportionate influence on the model.



The Actual vs Fitted plot confirms these issues: fitted values are negative for several low-count observations, which is not realistic for pedestrian counts, and the model underpredicts the highest observed values. These findings replicate the pattern seen in the First Trial modelling and motivate the use of a log-transformed response in the next section.

Log Model (along)

Variable Selection using Step Function

This step repeats the stepwise selection process on the log-transformed response, `log(obs_peak_along_ped_per_h)`, using the EDA-selected predictors. As with the linear model, both a null start and a full start are used to check whether the two approaches converge on a consistent set of predictors.

```
summary(log_step_both_null)
```

Model from Null Start

```
##
## Call:
## lm(formula = log(obs_peak_along_ped_per_h) ~ commercial_buildings +
##     education_low_year_3cat + footpath_both_sides, data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.00344 -0.41332  0.06658  0.50663  1.61991
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     1.9734     0.3083   6.400 8.70e-08 ***
## commercial_buildings            0.2933     0.0485   6.047 2.88e-07 ***
## education_low_year_3catKIN     -0.3755     0.3111  -1.207   0.2340
## education_low_year_3catY_plus   0.7425     0.4289   1.731   0.0904 .
## footpath_both_sidesTRUE         0.5033     0.2956   1.703   0.0957 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7849 on 44 degrees of freedom
## Multiple R-squared:  0.6711, Adjusted R-squared:  0.6412
## F-statistic: 22.45 on 4 and 44 DF,  p-value: 3.731e-10
```

`commercial_buildings`, `footpath_both_sides`, and `education_low_year_3cat` were selected. `commercial_buildings` has a strong, significant positive effect, consistent with the EDA. `footpath_both_sides` is marginally significant ($p = 0.096$), with a positive coefficient matching the EDA finding that pedestrian counts tend to be higher where footpaths exist on both sides. `education_low_year_3cat` shows a mixed pattern: the KIN level is not significant, while the Y_plus level is marginally significant ($p = 0.090$). The model explains about 64% of the variance (adjusted $R^2 = 0.641$).

```
summary(log_step_both_full)
```

Model from Full Start

```
##
## Call:
## lm(formula = log(obs_peak_along_ped_per_h) ~ commercial_buildings +
```

```
##     footpath_both_sides + education_low_year_3cat, data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.00344 -0.41332  0.06658  0.50663  1.61991
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     1.9734     0.3083   6.400 8.70e-08 ***
## commercial_buildings            0.2933     0.0485   6.047 2.88e-07 ***
## footpath_both_sidesTRUE         0.5033     0.2956   1.703   0.0957 .
## education_low_year_3catKIN     -0.3755     0.3111  -1.207   0.2340
## education_low_year_3catY_plus   0.7425     0.4289   1.731   0.0904 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7849 on 44 degrees of freedom
## Multiple R-squared:  0.6711, Adjusted R-squared:  0.6412
## F-statistic: 22.45 on 4 and 44 DF,  p-value: 3.731e-10
```

The full-start model selected exactly the same predictors as the null-start model: `commercial_buildings`, `footpath_both_sides`, and `education_low_year_3cat`.

Comparison of Stepwise Models As both starting points converged on the same model, this consistency supports the stability of the selected predictor set, and no further comparison is required before finalising the model.

Log Final Model

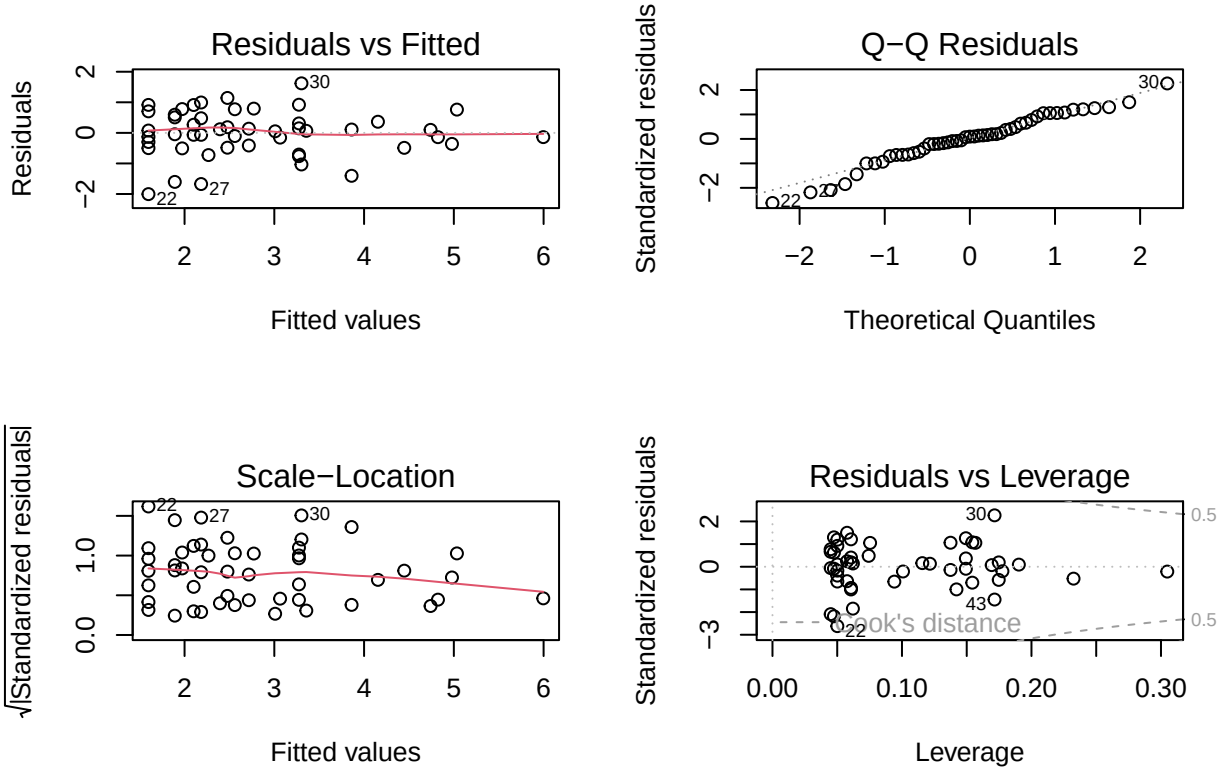
The final log model retains `commercial_buildings`, `footpath_both_sides`, and `education_low_year_3cat`, the same predictors selected by stepwise selection from both starting points.

```
### Log final model
log_final_model <- lm(log(obs_peak_along_ped_per_h) ~ commercial_buildings +
                        footpath_both_sides + education_low_year_3cat,
                      data = model_df)
summary(log_final_model)
```

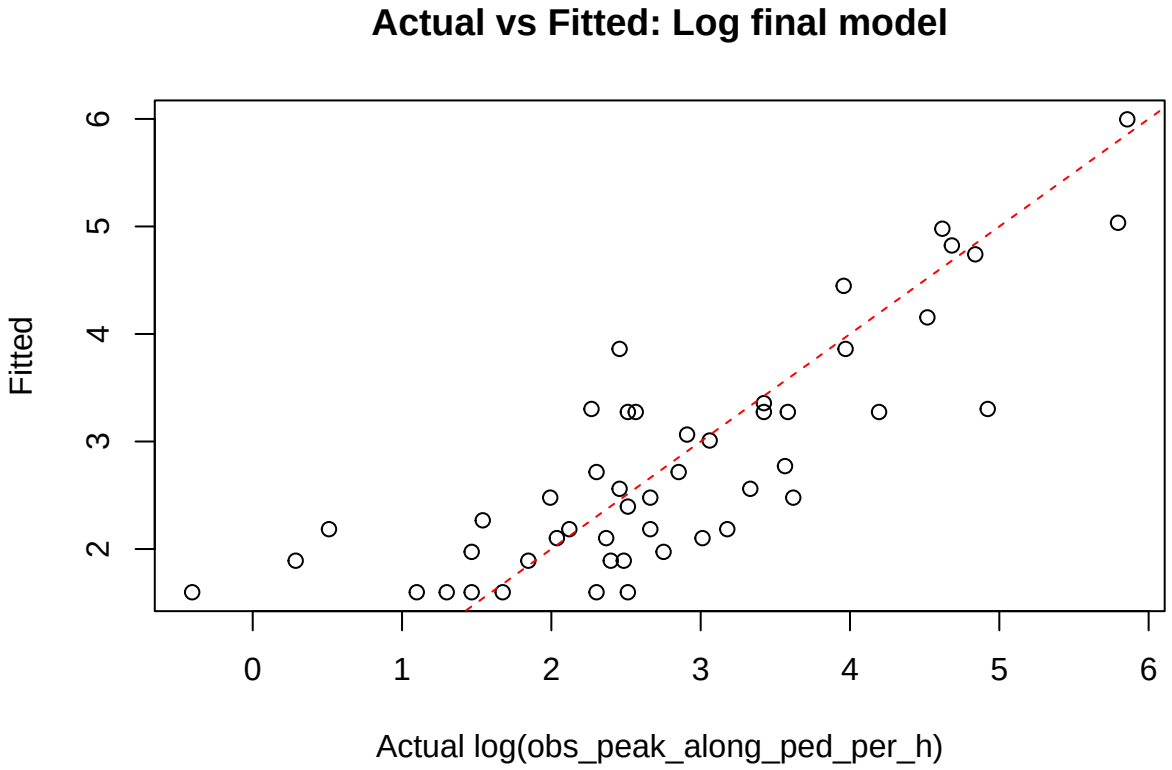
```
##
## Call:
## lm(formula = log(obs_peak_along_ped_per_h) ~ commercial_buildings +
##     footpath_both_sides + education_low_year_3cat, data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.00344 -0.41332  0.06658  0.50663  1.61991
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     1.9734     0.3083   6.400 8.70e-08 ***
## commercial_buildings            0.2933     0.0485   6.047 2.88e-07 ***
## footpath_both_sidesTRUE         0.5033     0.2956   1.703   0.0957 .
## education_low_year_3catKIN     -0.3755     0.3111  -1.207   0.2340
## education_low_year_3catY_plus   0.7425     0.4289   1.731   0.0904 .
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7849 on 44 degrees of freedom
## Multiple R-squared:  0.6711, Adjusted R-squared:  0.6412
## F-statistic: 22.45 on 4 and 44 DF,  p-value: 3.731e-10
```

`commercial_buildings` has a strong, significant positive effect, consistent with the EDA. `footpath_both_sides` is marginally significant (p = 0.096) with a positive coefficient, matching the EDA finding that pedestrian counts tend to be higher where footpaths exist on both sides. `education_low_year_3cat` shows a mixed pattern: the KIN level is not significant, while the Y_plus level is marginally significant (p = 0.090). The model explains about 64% of the variance (adjusted $R^2 = 0.641$).



The diagnostic plots show a clear improvement over the linear model. The Residuals vs Fitted plot shows no clear pattern, and the Scale-Location plot indicates roughly constant variance, suggesting that the heteroscedasticity seen in the linear model has been resolved. The Q-Q plot shows most points close to the line, with only point 30 as a mild outlier, compared to multiple outliers in the linear model. In the Residuals vs Leverage plot, point 30 has slightly elevated leverage but remains within Cook's distance, so it is not considered highly influential.



The Actual vs Fitted plot shows points distributed evenly along the diagonal, without the negative predictions or systematic underprediction seen in the linear model. These results are consistent with the First Trial findings, confirming that the log-transformed model provides a better fit than the linear model for `obs_peak_along_ped_per_h`.

Log Model (cross)

Variable Selection using Step Function

This step applies stepwise selection (both directions) to `log(obs_peak_cross_ped_per_h + 1)`, using the same EDA-selected predictors as for `along_ped`. A `+1` offset is applied because one observation has a value of zero, which would otherwise be undefined under a log transform. Both a null start and a full start are used to check whether the two approaches converge on a consistent set of predictors.

```
summary(log_cross_step_both_null)
```

Model from Null Start

```
##
## Call:
## lm(formula = log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings +
##     education_low_year_3cat + footpath_both_sides + park_bushland_reserve +
##     overture_grocery_convenience_pharmacy_count_500m + overture_other_shopping_count_500m,
##     data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.09736 -0.39131  0.03359  0.44317  1.68343
##
## Coefficients:
```

```
##                                             Estimate Std. Error t value
## (Intercept)                                  0.55273    0.57022   0.969
## commercial_buildings                         0.22053    0.06538   3.373
## education_low_year_3catKIN                   0.07651    0.40008   0.191
## education_low_year_3catY_plus                1.03907    0.50232   2.069
## footpath_both_sidesTRUE                      0.50972    0.34667   1.470
## park_bushland_reserveTRUE                    0.79324    0.39230   2.022
## overture_grocery_convenience_pharmacy_count_500m  0.15979    0.07305   2.187
## overture_other_shopping_count_500m          -0.03525    0.02498  -1.411
##                                             Pr(>|t|)
## (Intercept)                                  0.33806
## commercial_buildings                         0.00163 **
## education_low_year_3catKIN                   0.84929
## education_low_year_3catY_plus                0.04493 *
## footpath_both_sidesTRUE                      0.14911
## park_bushland_reserveTRUE                    0.04973 *
## overture_grocery_convenience_pharmacy_count_500m  0.03448 *
## overture_other_shopping_count_500m           0.16577
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9146 on 41 degrees of freedom
## Multiple R-squared:  0.6151, Adjusted R-squared:  0.5494
## F-statistic: 9.361 on 7 and 41 DF,  p-value: 7.071e-07
```

```
summary(log_cross_step_both_full)
```

Model from Full Start

```
##
## Call:
## lm(formula = log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings +
##     overture_grocery_convenience_pharmacy_count_500m + overture_other_shopping_count_500m +
##     footpath_both_sides + park_bushland_reserve + education_low_year_3cat,
##     data = model_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.09736 -0.39131  0.03359  0.44317  1.68343
##
## Coefficients:
##                                             Estimate Std. Error t value
## (Intercept)                                  0.55273    0.57022   0.969
## commercial_buildings                         0.22053    0.06538   3.373
## overture_grocery_convenience_pharmacy_count_500m  0.15979    0.07305   2.187
## overture_other_shopping_count_500m          -0.03525    0.02498  -1.411
## footpath_both_sidesTRUE                      0.50972    0.34667   1.470
## park_bushland_reserveTRUE                    0.79324    0.39230   2.022
## education_low_year_3catKIN                   0.07651    0.40008   0.191
## education_low_year_3catY_plus                1.03907    0.50232   2.069
##                                             Pr(>|t|)
## (Intercept)                                  0.33806
## commercial_buildings                         0.00163 **
## overture_grocery_convenience_pharmacy_count_500m  0.03448 *
```

```
## overture_other_shopping_count_500m                0.16577
## footpath_both_sidesTRUE                           0.14911
## park_bushland_reserveTRUE                         0.04973 *
## education_low_year_3catKIN                        0.84929
## education_low_year_3catY_plus                     0.04493 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9146 on 41 degrees of freedom
## Multiple R-squared:  0.6151, Adjusted R-squared:  0.5494
## F-statistic: 9.361 on 7 and 41 DF,  p-value: 7.071e-07
```

Comparison of Stepwise Models Stepwise selection converged on the same model regardless of the starting point (null or full), selecting `commercial_buildings`, `overture_grocery_convenience_pharmacy_count_500m`, `overture_other_shopping_count_500m`, `footpath_both_sides`, `park_bushland_reserve`, and `education_low_year_3cat`. This consistency supports the stability of the selected predictor set.

Log (Cross) Final Model

Based on the AIC and partial F-test comparisons, the additional predictors dropped during model comparison did not significantly improve model fit. The final log model for `cross_ped` retains three predictors: `commercial_buildings`, `overture_grocery_convenience_pharmacy_count_500m`, and `park_bushland_reserve`.

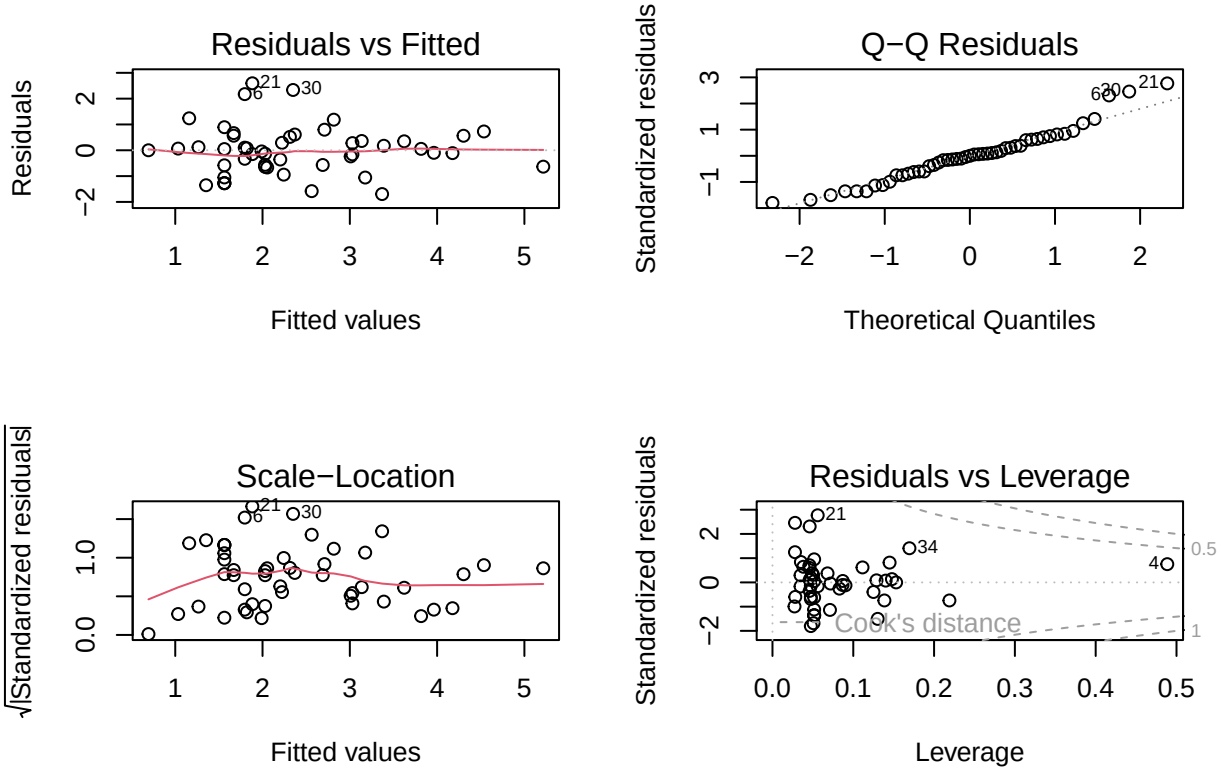
```
log_cross_final_model <- lm(log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings +
                              overture_grocery_convenience_pharmacy_count_500m +
                              park_bushland_reserve, data = model_df)
summary(log_cross_final_model)
```

```
##
## Call:
## lm(formula = log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings +
##     overture_grocery_convenience_pharmacy_count_500m + park_bushland_reserve,
##     data = model_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.6958 -0.5820  0.0001  0.5192  2.5943
##
## Coefficients:
##                                                Estimate Std. Error t value
## (Intercept)                                     0.69304    0.37721   1.837
## commercial_buildings                            0.23329    0.05705   4.089
## overture_grocery_convenience_pharmacy_count_500m  0.10674    0.05779   1.847
## park_bushland_reserveTRUE                       0.86975    0.38251   2.274
##                                                Pr(>|t|)
## (Intercept)                                    0.072777 .
## commercial_buildings                           0.000176 ***
## overture_grocery_convenience_pharmacy_count_500m 0.071317 .
## park_bushland_reserveTRUE                      0.027798 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9642 on 45 degrees of freedom
## Multiple R-squared:  0.5305, Adjusted R-squared:  0.4992
```

```
## F-statistic: 16.95 on 3 and 45 DF,  p-value: 1.654e-07
```

`commercial_buildings` has a strong, significant positive effect, consistent with the EDA. `park_bushland_reserve` is significant ($p = 0.028$) with a positive coefficient, also matching the EDA finding. `overture_grocery_convenience_pharmacy` is marginally significant ($p = 0.071$), with a positive coefficient in the expected direction. The model explains about 50% of the variance (adjusted $R^2 = 0.499$), somewhat lower than the along_ped model, reflecting the weaker overall relationships found for cross_ped in the EDA.

```
par(mfrow = c(2, 2))
plot(log_cross_final_model)
```

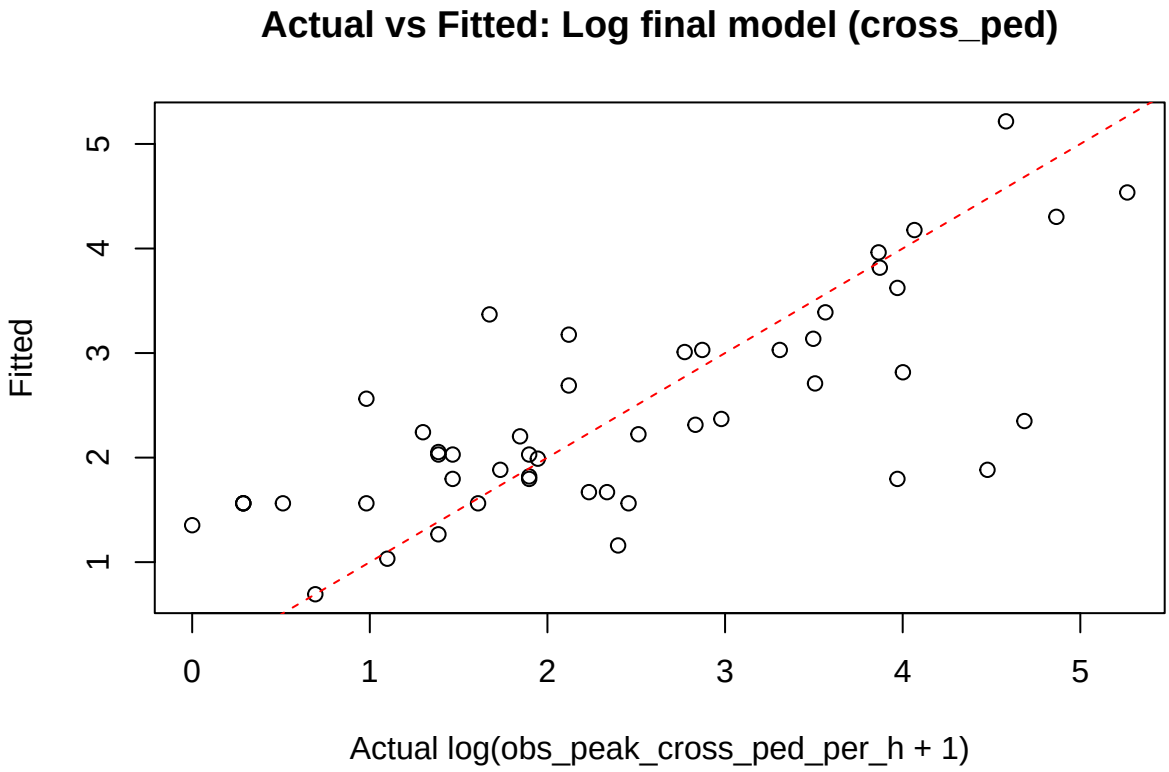



```
par(mfrow = c(1, 1))
```

The diagnostic plots show broadly acceptable behaviour, though with more residual spread than the along_ped model. The Residuals vs Fitted plot shows no strong pattern, though points 6, 21, and 30 stand out with larger residuals. The Scale-Location plot indicates roughly constant variance across the fitted range. The Q-Q plot shows most points close to the line, with some deviation at the upper tail (points 6, 21, and 30). In the Residuals vs Leverage plot, points 4 and 34 show relatively high leverage, but none exceed Cook's distance, so no observation appears to be highly influential.

```
fitted_vals <- fitted(log_cross_final_model)
actual_vals <- log(model_df$obs_peak_cross_ped_per_h + 1)

plot(actual_vals, fitted_vals,
     xlab = "Actual log(obs_peak_cross_ped_per_h + 1)",
     ylab = "Fitted",
     main = "Actual vs Fitted: Log final model (cross_ped)")
abline(0, 1, col = "red", lty = 2)
```



The Actual vs Fitted plot shows a reasonable linear relationship along the diagonal, though with more scatter than the along_ped model, particularly at higher observed values. This is consistent with the lower adjusted R^2 and reflects the comparatively weaker predictive relationships identified for `cross_ped` during the EDA.

Models Considered but Not Adopted

A negative binomial model was also considered for the count response. However, the observed values are multi-day averages rather than integer counts, so fitting it would have required rounding, which distorts low-count sites. It was therefore not adopted as a formal comparison candidate.

Back-Transformation and Smearing Estimator

Both log models predict on the log scale. Back-transforming with $\exp(\hat{\mu})$ estimates the geometric, not arithmetic, mean of the response, so the back-transformed predictions systematically under-predict (Duan, 1983). Duan's smearing estimator corrects this with a single multiplicative factor estimated from the residuals $\hat{\varepsilon}_i$ of the log model:

$$\hat{S} = \frac{1}{n} \sum_{i=1}^{n} \exp(\hat{\varepsilon}_i)$$

The corrected predictions are $\hat{y} = \exp(\hat{\mu})\,\hat{S}$ for the along model and $\hat{y} = \exp(\hat{\mu})\,\hat{S} - 1$ for the cross model, where the -1 reverses the $+1$ offset.

Table 2: Smearing factors and mean predictions on the training data (pedestrians per hour)

Model	Smearing factor	Observed mean	Naive mean	Smeared mean
Log (along)	1.278	39.871	34.792	44.480

Model	Smearing factor	Observed mean	Naive mean	Smeared mean
Log (cross)	1.684	25.388	18.851	32.434

For the along model, the smearing factor is $\hat{S} = 1.278$. On the training data, the naive back-transformed predictions under-predict the observed mean by about 13%. Because the bias is multiplicative, its absolute size grows with the predicted count, so the busiest sites are affected most. For the cross model, the smearing factor is 1.684.

These values are in-sample and are not comparable with the grouped cross-validation results of the team benchmark. The correction is documented here rather than built into the deployed model: it should be applied once pedestrian counts from new sites confirm that the uncorrected model under-predicts.

Appendix

Log (along) Final Model Analysis

```
# Four candidate models
log_m1 <- lm(log(obs_peak_along_ped_per_h) ~ commercial_buildings +
               footpath_both_sides + education_low_year_3cat, data = model_df)
log_m2 <- lm(log(obs_peak_along_ped_per_h) ~ commercial_buildings +
               footpath_both_sides, data = model_df)
log_m3 <- lm(log(obs_peak_along_ped_per_h) ~ commercial_buildings +
               education_low_year_3cat, data = model_df)
log_m4 <- lm(log(obs_peak_along_ped_per_h) ~ commercial_buildings, data = model_df)
```

```
# AIC comparison across all four
AIC(log_m1, log_m2, log_m3, log_m4)
```

```
##        df      AIC
## log_m1  6 122.0480
## log_m2  4 128.5715
## log_m3  5 123.1743
## log_m4  3 127.8629
```

```
# Partial F-tests for nested comparisons
anova(log_m4, log_m2)  # does footpath_both_sides add value over commercial_buildings alone?
```

```
## Analysis of Variance Table
##
## Model 1: log(obs_peak_along_ped_per_h) ~ commercial_buildings
## Model 2: log(obs_peak_along_ped_per_h) ~ commercial_buildings + footpath_both_sides
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     47 34.499
## 2     46 33.602  1   0.89735 1.2284 0.2735
```

```
anova(log_m4, log_m3)  # does education_low_year_3cat add value over commercial_buildings alone?
```

```
## Analysis of Variance Table
##
## Model 1: log(obs_peak_along_ped_per_h) ~ commercial_buildings
## Model 2: log(obs_peak_along_ped_per_h) ~ commercial_buildings + education_low_year_3cat
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1     47 34.499
## 2     45 28.894  2    5.6056 4.3652 0.01851 *
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(log_m2, log_m1)  # does education_low_year_3cat add value once footpath is included?
```

```
## Analysis of Variance Table
##
## Model 1: log(obs_peak_along_ped_per_h) ~ commercial_buildings + footpath_both_sides
## Model 2: log(obs_peak_along_ped_per_h) ~ commercial_buildings + footpath_both_sides +
##     education_low_year_3cat
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
## 1     46 33.602
## 2     44 27.108  2    6.4942 5.2705 0.008872 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(log_m3, log_m1)  # does footpath_both_sides add value once education is included?
```

```
## Analysis of Variance Table
##
## Model 1: log(obs_peak_along_ped_per_h) ~ commercial_buildings + education_low_year_3cat
## Model 2: log(obs_peak_along_ped_per_h) ~ commercial_buildings + footpath_both_sides +
##     education_low_year_3cat
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     45 28.894
## 2     44 27.108  1    1.7859 2.8988 0.0957 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Log (cross) Final Model Analysis

```
# Full stepwise model (6 predictors)
log_cross_m1 <- lm(log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings +
                     overture_grocery_convenience_pharmacy_count_500m +
                     overture_other_shopping_count_500m +
                     footpath_both_sides + park_bushland_reserve +
                     education_low_year_3cat, data = model_df)

# Drop overture_other_shopping_count_500m (weakest, p = 0.166)
log_cross_m2 <- lm(log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings +
                     overture_grocery_convenience_pharmacy_count_500m +
                     footpath_both_sides + park_bushland_reserve +
                     education_low_year_3cat, data = model_df)

# Also drop footpath_both_sides
log_cross_m3 <- lm(log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings +
                     overture_grocery_convenience_pharmacy_count_500m +
                     park_bushland_reserve + education_low_year_3cat, data = model_df)

# Also drop education_low_year_3cat
log_cross_m4 <- lm(log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings +
                     overture_grocery_convenience_pharmacy_count_500m +
                     park_bushland_reserve, data = model_df)
```

```
# AIC comparison across all four candidates
AIC(log_cross_m1, log_cross_m2, log_cross_m3, log_cross_m4)
```

```
##              df      AIC
## log_cross_m1  9 139.5707
## log_cross_m2  8 139.8943
## log_cross_m3  7 140.2063
## log_cross_m4  5 141.3133
```

```
# Partial F-tests for nested comparisons
anova(log_cross_m2, log_cross_m1)  # does overture_other_shopping_count_500m add value?
```

```
## Analysis of Variance Table
##
## Model 1: log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings + overture_grocery_convenience_pha
##     footpath_both_sides + park_bushland_reserve + education_low_year_3cat
## Model 2: log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings + overture_grocery_convenience_pha
##     overture_other_shopping_count_500m + footpath_both_sides +
##     park_bushland_reserve + education_low_year_3cat
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     42 35.960
## 2     41 34.294  1    1.6655 1.9911 0.1658
```

```
anova(log_cross_m3, log_cross_m2)  # does footpath_both_sides add value?
```

```
## Analysis of Variance Table
##
## Model 1: log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings + overture_grocery_convenience_pha
##     park_bushland_reserve + education_low_year_3cat
## Model 2: log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings + overture_grocery_convenience_pha
##     footpath_both_sides + park_bushland_reserve + education_low_year_3cat
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     43 37.697
## 2     42 35.960  1    1.7373 2.0291 0.1617
```

```
anova(log_cross_m4, log_cross_m3)  # does education_low_year_3cat add value?
```

```
## Analysis of Variance Table
##
## Model 1: log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings + overture_grocery_convenience_pha
##     park_bushland_reserve
## Model 2: log(obs_peak_cross_ped_per_h + 1) ~ commercial_buildings + overture_grocery_convenience_pha
##     park_bushland_reserve + education_low_year_3cat
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     45 41.838
## 2     43 37.697  2     4.141 2.3618 0.1064
```

Multicollinearity Check (VIF)

Variance inflation factors are computed for the three final models with `car::vif()`. For factor predictors, the generalised VIF (GVIF) is reported with its adjusted value $\text{GVIF}^{1/(2\,\text{Df})}$, which is comparable with $\sqrt{\text{VIF}}$ for a numeric predictor. For numeric predictors, GVIF equals the ordinary VIF.

```
## Loading required package: carData
```

Table 3: Variance inflation factors of the final models

Model	Term	GVIF	Df	GVIF^(1/(2*Df))
Linear (along)	commercial_buildings	2.19	1	1.48
Linear (along)	parking_areas	2.19	1	1.48

Model	Term	GVIF	Df	GVIF^(1/(2*Df))
Log (along)	commercial_buildings	1.64	1	1.28
Log (along)	footpath_both_sides	1.65	1	1.28
Log (along)	education_low_year_3cat	1.05	2	1.01
Log (cross)	commercial_buildings	1.50	1	1.22
Log (cross)	overture_grocery_convenience_pharmacy_count_500m	1.47	1	1.21
Log (cross)	park_bushland_reserve	1.05	1	1.03

The largest GVIF across the three final models is 2.19. All VIFs are below 5, so multicollinearity is not a serious concern for these models. A low VIF does not, however, allow the negative coefficient of `parking_areas` in the linear model to be read causally: `parking_areas` is correlated with `commercial_buildings` ($r = 0.74$), so its coefficient reflects the effect conditional on commercial density and may be confounded.

References

Duan, N. (1983). Smearing estimate: A nonparametric retransformation method. *Journal of the American Statistical Association*, 78(383), 605–610.